

Output Ontology (OO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Justification

Output ontology is well-aligned with the deployment. Ground-truth labels are single correct answers per MCQ drawn from exam answer keys [Q12], which exactly matches the deployment's binary correct/incorrect requirement with positive/negative marking. The label schema is binary at inference time, and the elicitation explicitly confirmed that 'evaluation schemas across boards and states are described as broadly similar' [A4], so competitive-exam keyed answers are considered compatible. The 41-subject 8-domain organisation [Q42] supports per-subject reporting useful for diagnostic analysis. Domain-level analysis surfaces a known weakness of LLMs in culturally specific areas [Q20, Q71, Q77], which the deployment user should be aware of. The taxonomy was derived bottom-up from ~20K fine-grained tags clustered and manually merged [Q39, Q40, Q41], which is sound methodology, although whether the resulting categories carry any culturally variable correct-answer norms is not analysed.

Strengths

- Single-correct-answer ground truth from exam answer keys directly matches binary scoring requirement [Q12]
- 41-subject 8-domain organisation supports diagnostic per-subject reporting [Q42]
- Cross-language answer-key consistency for parallel items [DATASET-D18, D19] confirms reliable label application

ID	Evidence
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूरगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check

- Elicitation confirms competitive-exam keys are broadly compatible with North Indian board norms [A4]

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are MCQ option indices (option1–option4) with a single target [Q12, DATASET-D1]. This is regionally relevant — Indian competitive and board MCQs use the same single-correct format with positive/negative marking [WEB-1, WEB-2].

ID	Evidence
WEB-1	Indian competitive exams use single-correct MCQ with positive/negative marking — taxonomically aligned with MILU output schema
WEB-2	UPSC and similar exams confirm single-answer MCQ output convention

OO-2: No regionally specific output categories appear missing for a binary MCQ deployment. The deployment does not require partial-credit, rubric, or multi-label output [A1, A2].

OO-3: No categories visibly encode non-regional values. The user confirmed evaluation schemas are broadly consistent across North Indian boards and competitive exams [A4], reducing the risk that competitive-exam answer keys diverge from board norms.

OO-4: Stakeholder-driven taxonomy redesign is not required — the deployment uses the same single-answer MCQ taxonomy as the benchmark.

OO-5: No major taxonomy issues identified. Minor caveat: domain-level categorisation may aggregate over heterogeneous content (e.g., Arts & Humanities mixing competitive-GK and literature) [Q20, Q77], which could affect interpretation of domain-level scores.

Information Gaps

- Whether domain-level aggregation hides subject-level mismatches relevant to the teacher deployment

Requires Expert Verification

- Whether competitive-exam answer keys for items in Hindi Literature and Civics agree with state-board mark-scheme conventions in detail